

Econometrics III
Lindo, Fall 2013
Assignment 7
Due: November 6

The usual note applies. Please copy and paste the final table of results into your do file (commented out) where appropriate so that the code can be—but need not be—run to see the results of your work.

In this problem set, you will use data from the Panel Study of Income Dynamics, 1968-2007, to estimate the effects of job displacements on individuals' earnings, as in Stevens (1997) and Lindo (2010). The PSID is a longitudinal data set that tracks households over time (and adds new households to the sample when new households are formed). One complication of using the PSID is that the data contain multiple observations for each household in each year and includes many "family-level variables" corresponding to characteristics of the household, characteristics of the head, and characteristics of the "wife." However, I've simplified things for you by providing a dataset that has been limited to males between age 23 and 65 who are defined as the "head of household." In order to estimate the effect of job displacements, you should proceed as follows:

- Individuals are uniquely identified by their `intnum68` and `seqnum68`. Use the `egen var=group()` command to create a single identifier to simplify things in the next step.
- Identify observations in which the individual has changed jobs in the last year because of a layoff or firm closure. You'll use several variables here because there is a variable for individuals who are currently employed, unemployed, etc. In particular, use `hpnd`, `hpnde`, `hpndu`, `hpndnolf`, `hpndotr`. If any of these variables takes a value of 3, the individual experienced a layoff. If any of these variables takes a one, the individual experienced a firm closure.
- Drop any individuals who have a displacement identified in 1968. For these folks, the question asked about "the last 10 years" so we don't know how recent it was and won't be able to construct leads and lags appropriately.
- Construct a set of displacement indicators corresponding to 3 years prior to first displacement, 2 years prior to first displacement, ..., 8+ years following first displacement. Assume that the job displacement happened the year before it is identified in the data. (Stevens 1997 shows that only first displacements should be considered (conditionally) exogenous since initial displacements increase the risk of future displacements.)
- Create a full set of year indicator variables, a quadratic in age, and log family income (using `inc` and dropping observation where the value has been imputed, as indicated by "`acc-fuminc!=0`").
- Estimate the coefficients on the set of displacement indicators using 6 models. 1. Pooled OLS regression controlling for year fixed effects and a quadratic in age assuming homoskedasticity. 2. Same but with standard error estimates clustered on the individual. 3. Individual fixed effects regression controlling for year fixed effects and a quadratic in age assuming homoskedasticity. 4. Same but with standard error estimates clustered on the individual. 5. Individual fixed effects regression controlling for year fixed effects and a quadratic in age assuming homoskedasticity with a balanced panel such that no observations 9+ years after displacement are included. 6. Same but with standard error estimates clustered on the individual. *You should store the regression estimates in memory using the `eststo` command and then display them (after all have been collected) using the `esttab` command. Be sure to use the options so that the standard errors are displayed and *only* the coefficient estimates on the displacement indicators are displayed.*

Questions to be answered within your do file (commented out):

1. What does the pooled OLS estimate tell you about selection into job displacements?
2. Time and time again, studies have found that earnings begin to fall below their expected levels before firms shut down. Is this a threat to the validity of our preferred models (4 and 6)? Refer to the estimates as you see fit and explain.

Submit the do-file your TA electronically. If it does not run on our computers unaltered, it will be returned to you for revision. In order to facilitate this, include the following at the top of your code:

```
capture cd "<insert what you want to use as your working directory for this assignment here>"
if c(username)=="apackham" {
    cd "C:\Documents and Settings\apackham\Desktop\ECMT677\Assignments\Assignment 7 - PSID Displacements"
}
```